

Waterproofing Materials

Introduction

Water proofing

Protection to prevent water entering in internal and external building structures like toilets, swimming pools, water tanks, retaining walls, roofs etc.

Damp proofing

Treatment to a surface to resist the passage of water in areas like foundations, exterior walls, metal surfaces, metal staircases etc.

Common Water Damage problems

- Corrosion of steel reinforcement in concrete
- Blistering of paint
- Dampness on walls, roofs & Floors
- Leakage in water tanks



Benefits of Waterproofing

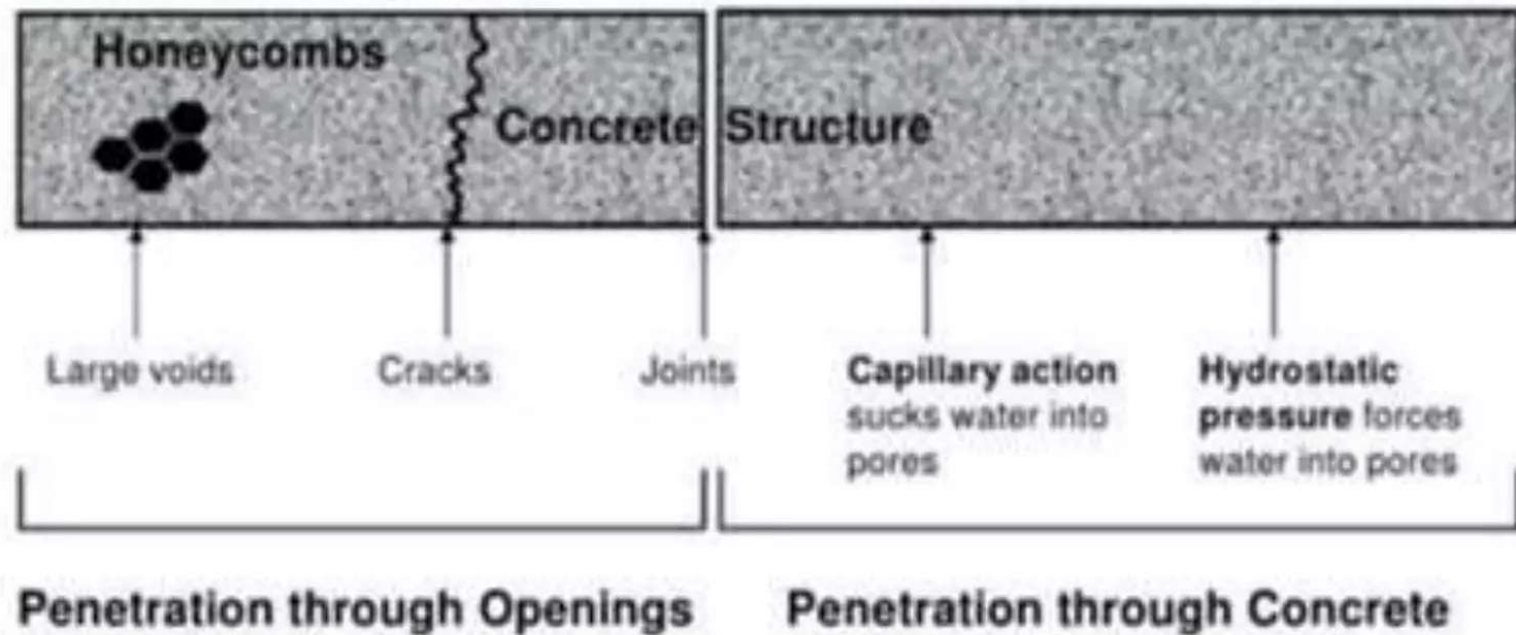
- Increased durability of concrete structures
- Reduce chemical attack
- Cost reduction in maintenance
- Prevents growth of dangerous mold

Water proofing may also be done after a building is built ,to address problems as they emerge or as part of a building retrofit. Water proofing is done in various parts of the building which include

- WC
- Bathrooms
- Terrace
- Roofs and chajjas
- Basement
- Swimming pools
- Underground ducts
- Underground and over head water tanks

Water proofing maintains the appearance of the building and increases the life of the structure

How Water can penetrate Concrete Structures?



Why Waterproofing is Important

- When we refer back to the definition of waterproofing as a barrier to prevent water entering or escaping from structure, then the real importance is reflected in the consequences of not waterproofing.

What could happen if a building isn't waterproof ?

- Poor waterproofing brings serious risk to the integrity of a building.
- When exposed to water ,foundation cracks or joints may lead to bigger structural problems. These are, Leakage ,Spalling ,Deterioration
- If the building is made up of wood or has wood furnishings the moisture from water intrusion will cause wood to rot or delaminate.
- Mould spores also pose hazards. Exposure to these may lead to allergies, asthma, irritations and fungal infection.

Waterproofing Methods

- Cementitious Waterproofing
- Liquid Waterproofing Membrane
- Bituminous Membrane
- Bituminous Coating
- Polyurethane Liquid Membrane

1. Cementitious Waterproofing

- Cementitious waterproofing is the easiest method of waterproofing in construction. The materials for cementitious waterproofing are readily available from suppliers of masonry products. And they are easy to mix and apply.
- The applications of the cementitious waterproofing technique are in the internal wet areas, such as toilets. That is why it does not go through the contract and expansion process.

Cementitious Waterproofing

- a. Materials – Masonry products
- b. This method is often used in the internal wet area.
 - Toilets
 - Laundries
 - Kitchens
 - Entries
 - This method is semi-flexible or rigid.
 - It is not exposed to sunlight and weathering.

Cementitious Waterproofing cont.



Applications of Cementitious Waterproofing

- Water treatment plants
- Sewage treatment facilities
- Bridges
- Dams
- Railway and subway systems
- Marine cargo ports and docks
- River locks/channels
- Parking structures
- Tunnels

2. Liquid Waterproofing Membrane

- The liquid membrane consists of a primer coat and two topcoats. The application of the coatings is by spray, roller, or trowel. The liquid layer is thin and offers more flexibility than the cementitious types of waterproofing.
- The liquid cures into a rubbery coating on the wall. The elongation properties of the coat can reach as high as 280%. The durability of the waterproofing coating depends on what type of polymer the manufacturer used for the making of the liquid waterproofing.

Liquid Waterproofing Membrane cont.



Liquid waterproofing membrane can be of a spray-applied liquid layer composed of polymer-modified asphalt. Polyurethane liquid membranes in separate grades for trowel, roller, or spray are also available from various manufacturers.

Liquid Waterproofing Membrane cont.

Equipment used for coatings-

- Spray
- Roller
- trowel
- Which may be aqueous solution or gas mixture.
- Liquid membrane should be immiscible with water.
- Low vapor pressure.
- This methods more flexible.

Advantage

- A liquid waterproofing membrane is essential for **protecting many building structures**. These flexible membranes are hand or spray applied onto surfaces such as balconies, podium decks and green roofs. They are even used to line municipal water tanks to prevent leaks.

Cont.

Liquid Waterproofing Membrane is used in the following type of structures:

- Balconies
- Podium decks
- Green roofs
- Water tank

3.Bituminous Coating Waterproofing

- Bituminous coating (asphalt coating) is made of bitumen based materials. It is a flexible protective coat based on its formulation and polymerization grade. The flexibility and protection against water can be influenced by the polymer grade and reinforcement of fiber.
- The most common applications of bituminous coatings include areas that are beneath screed wet. It is an excellent protective coating and waterproofing agent, especially on surfaces such as concrete foundations.
- It is not suitable for exposure to sunlight unless it is modified with a more flexible material such as polyurethane or acrylic-based polymers.

Bituminous Coating Waterproofing cont.



Bituminous Coating cont.

- Bituminous Coating is also called as asphalt coating.
- This methods commonly use beneath screed wet area.
- It is an excellent waterproofing methods for a concrete foundation.
- It is made of bitumen based materials.
- Should not expose to sunlight.
- Exposure to the sunlight can be unstable and fragile.

Bituminous Membrane cont.

- Its popular method used for low-sloped roofs.
- Bituminous waterproofing membrane have torch on membrane and self-adhesive membrane.
- Self-adhesive compounds-
 - comprise asphalt
 - Polymers
 - Filler
- Certain resins and oils may be added to improve adhesion characteristics.

Bituminous Membrane Waterproofing cont.



5. Polyurethane Liquid Membrane Waterproofing

- Polyurethane liquid membrane method of waterproofing is used for the flat roof area and exposed to weathering. This waterproofing method is expensive.

Polyurethane Liquid Membrane Waterproofing cont.



Polyurethane Liquid Membrane can offer higher flexibility. Polyurethane is very sensitive to moisture content. Therefore before application, one has to be very careful evaluating the moisture content of the concrete slab, otherwise peeling or de-bonding of membranes may happen after some time

Polyurethane Liquid Waterproofing Membrane

Equipment used for coatings-

- Spray
- Roller
- trowel
- Which may be aqueous solution or gas mixture.
- Liquid membrane should be immiscible with water.
- Low vapor pressure.
- This methods more flexible.

Advantages of Polyurethane liquid waterproofing membranes

- No significant load increase on lightweight decks and roofs.
- UV resistant, no coat protection required.
- Easy to repair, even locally. long term maintenance is quick and low cost

Thank You

